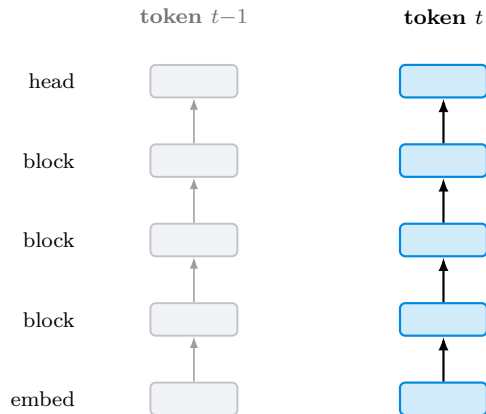


Inference-time information flows

Depth recurrence OFF

Vanilla transformer

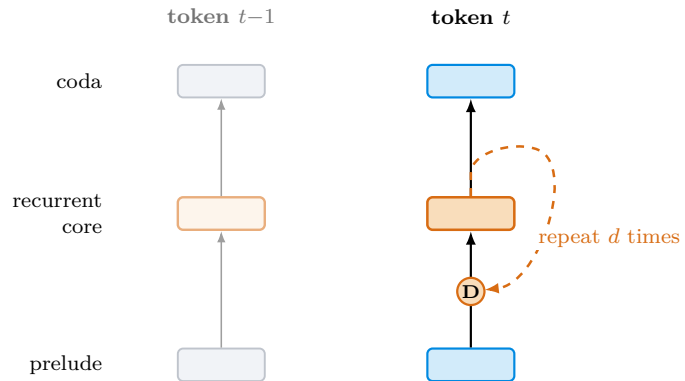
Each token is processed once; there is no temporal state and no depth loop.



Depth recurrence ON

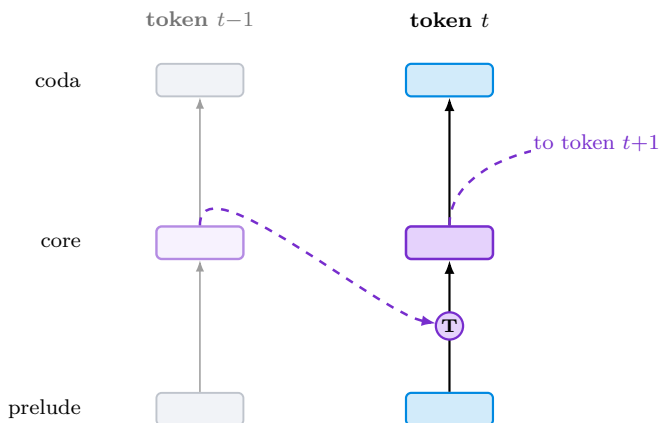
Depth-recurrent / looped

Depth state is local to each token; the shared recurrent core is reused several times within token t .



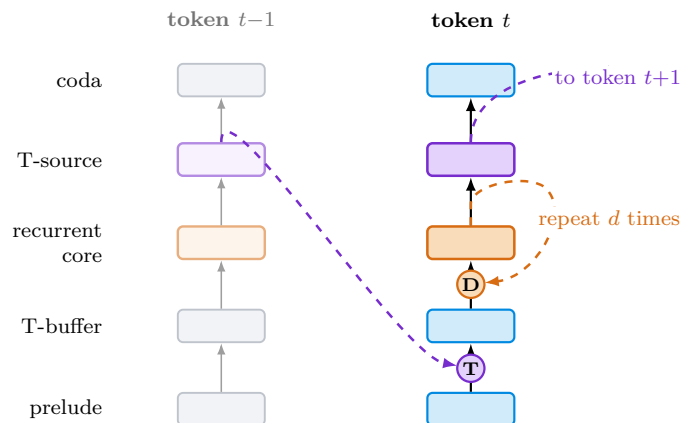
Temporally recurrent

A temporal hidden state from token $t-1$ is read while processing token t .



Depth + temporal hybrid

Temporal state crosses tokens; depth state is reused only within token t .



Temporal recurrence OFF

Temporal recurrence ON